

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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Name	Donthu Raghava
Register Number	192525304

MODEL PRACTICAL EXAMINATION – SET 4

Answer all questions. Each question carries equal marks (20).

Total marks: 100

1. Create a simple cloud software application to run a C program to display the student information using any Cloud Service Provider to demonstrate SaaS, with the template of Student Name, Register No., Address, Phone, Age, Courses, Grades and Attendance Report, Progress Report, Semester Mark Sheet forms.
2. Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Student, Register No., Subjects, Marks, Average, Percentage, Rank, Overall Performance, etc.
3. Create a Clone of a VM and test it by loading the previous version/cloned VM.
4. Install a C compiler in the virtual machine created using VirtualBox/VMware Workstation/Player and execute simple C programs.

Experiment 1: Create a simple cloud software application to run a C program to display the student information using any Cloud Service Provider to demonstrate SaaS, with the template of Student Name, Register No., Address, Phone, Age, Courses, Grades and Attendance Report, Progress Report, Semester Mark Sheet forms.

Aim:

To create a simple cloud software application for Student Information Management System using Zoho Creator to demonstrate Software as a Service (SaaS).

Procedure:

Step 1: Go to www.zoho.com.

Step 2: Log in to your Zoho account.

Step 3: Open **Zoho Creator** and select **Create New Application**.

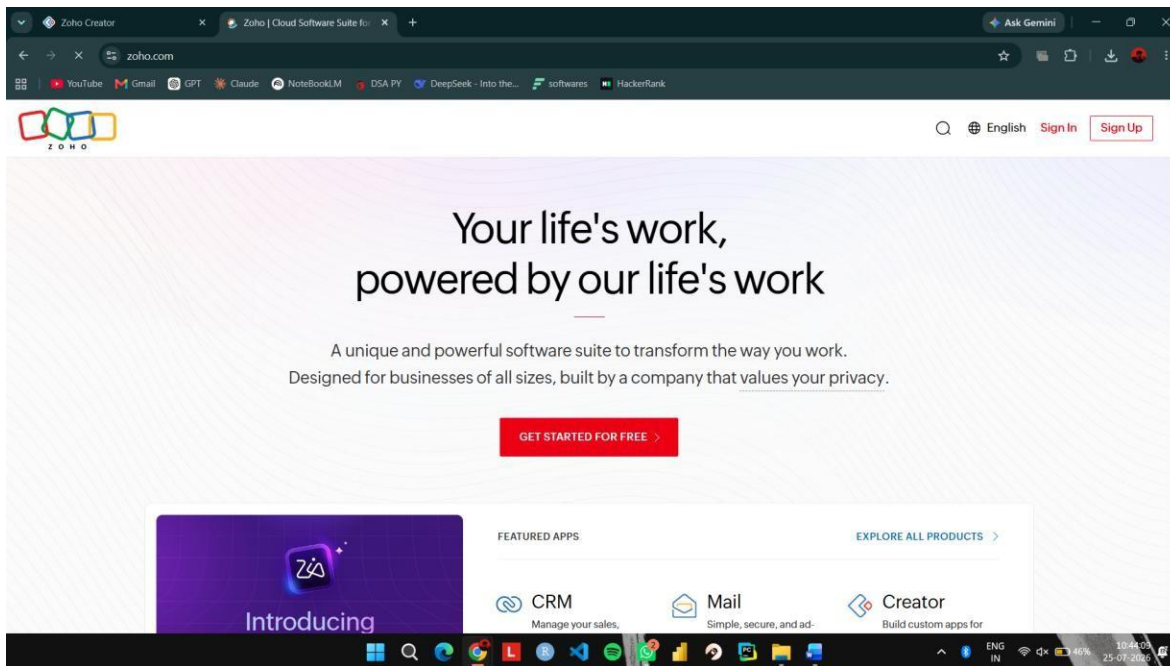
Step 4: Enter the application name as **Student Information Management System**.

Step 5: Create the required forms, reports, dashboard, and C Program Submission form.

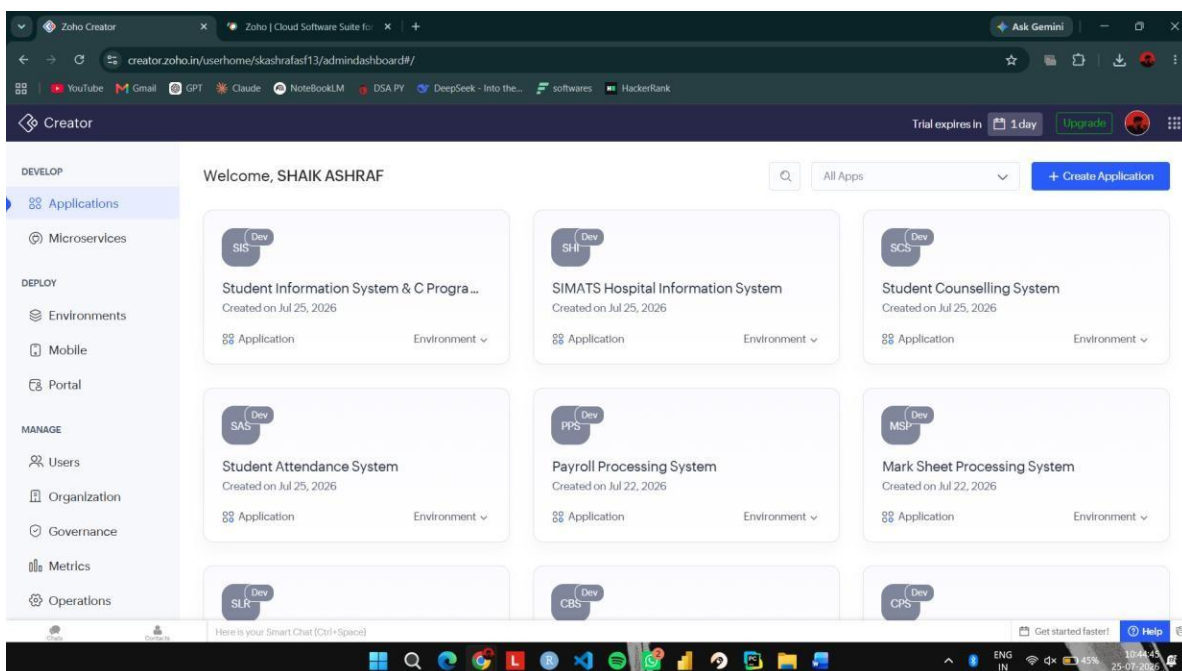
Step 6: Save and publish the application. The software has been created successfully.

IMPLEMENTATION:

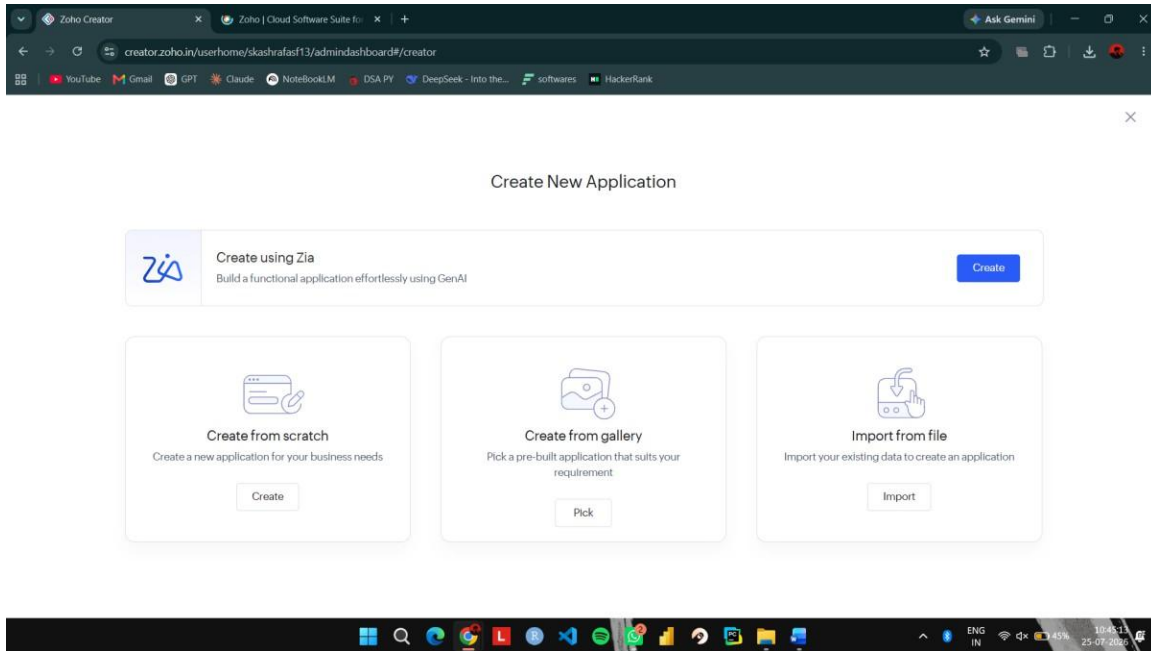
Step 1: Go to www.zoho.com.



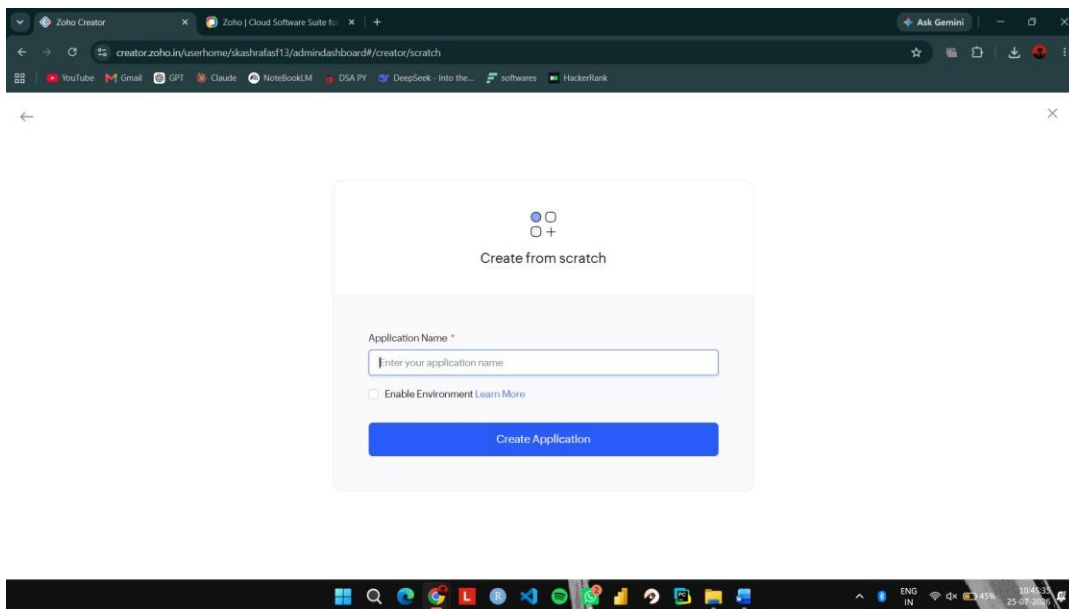
Step 2: Log in to your Zoho account.



Step 3: Open Zoho Creator and select Create New Application.



Step 4: Enter the application name as Student Information Management System.



Step 5: Create the required forms, reports, dashboard, and C Program Submission form.

The screenshot shows the Zoho Creator form builder interface for a form titled "C Program Submission". The interface is divided into three main sections: "Basic Fields", "Field Properties", and a "Done" button.

Basic Fields: This section contains a list of fields that can be added to the form. The fields are:

- Name
- Email
- Address
- Phone
- Single Line
- Multi Line
- Number
- Date
- Time
- Drop Down

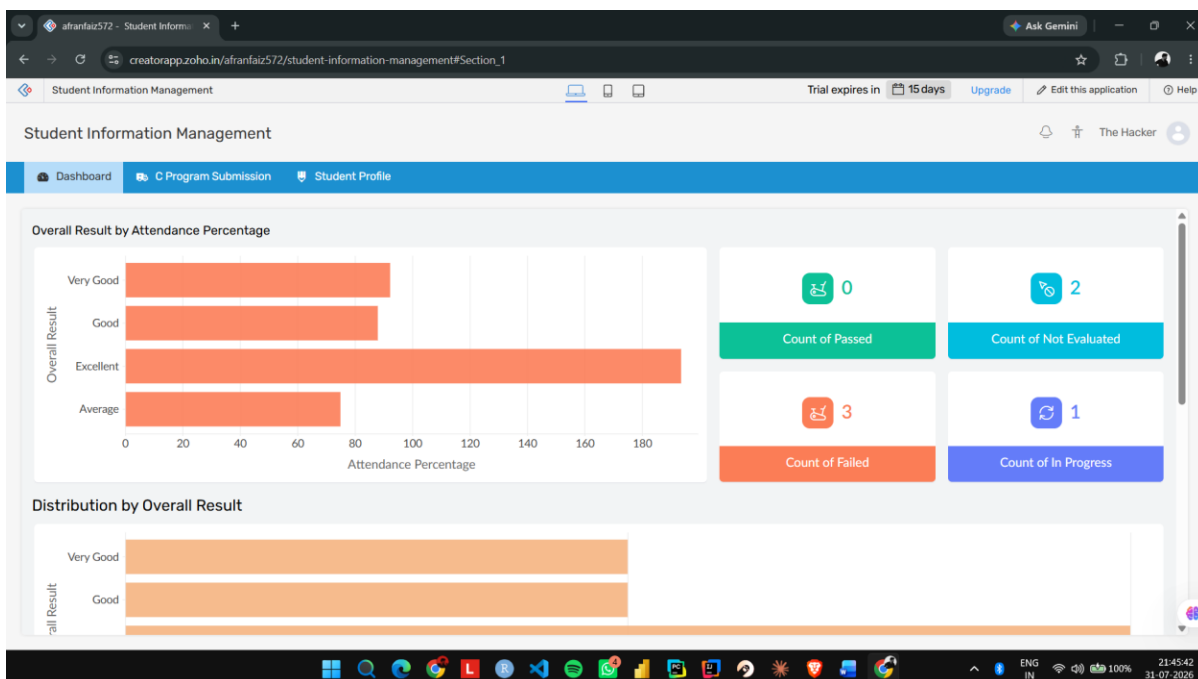
The "Student" field is selected, and its properties are displayed in the "Field Properties" section on the right.

Field Properties:

- Field name: Student
- Field link name: student
- View Field References
- Display Fields: Student Information Management → Student Profile
- Display type: Multi Select
- Validation: Mandatory

The "Done" button is located at the top right of the interface.

Step 6: Save and publish the application. The software has been created successfully.



Experiment 2: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Student, Register No., Subjects, Marks, Average, Percentage, Rank, Overall Performance, etc.

Aim:

To create a simple cloud software application for Mark Sheet Processing System using Zoho Creator to demonstrate Software as a Service (SaaS).

Procedure:

Step 1: Go to www.zoho.com.

Step 2: Log in to your Zoho account.

Step 3: Open **Zoho Creator** and select **Create New Application**.

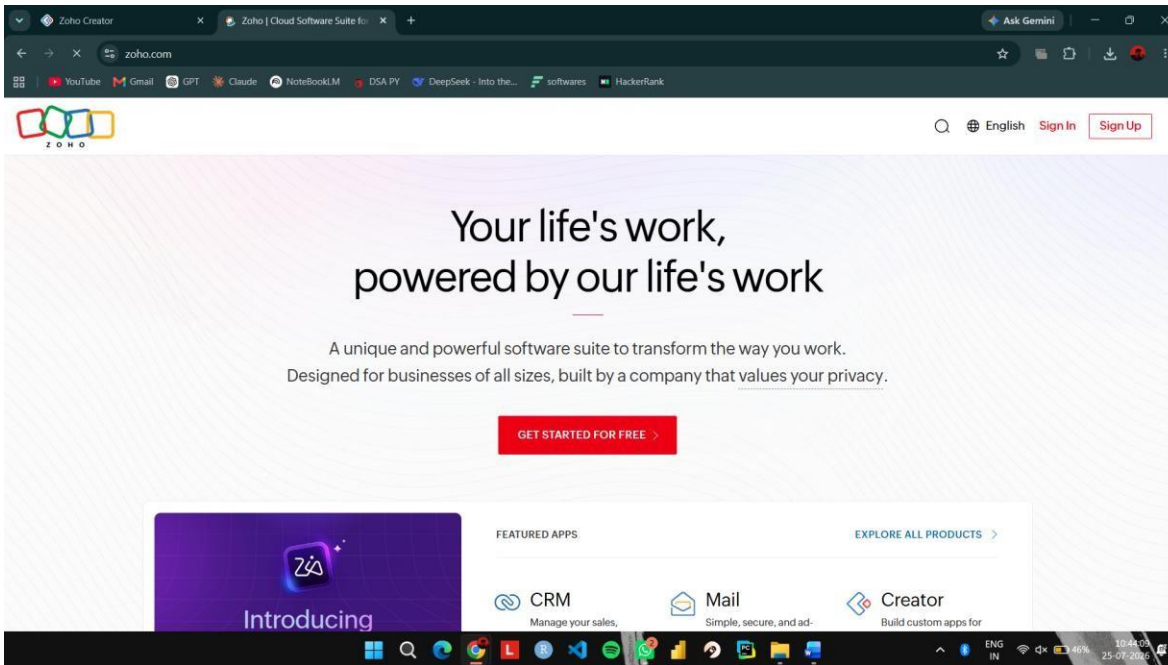
Step 4: Enter the application name as **Mark Sheet Processing System**.

Step 5: Create the required forms, reports, and dashboard.

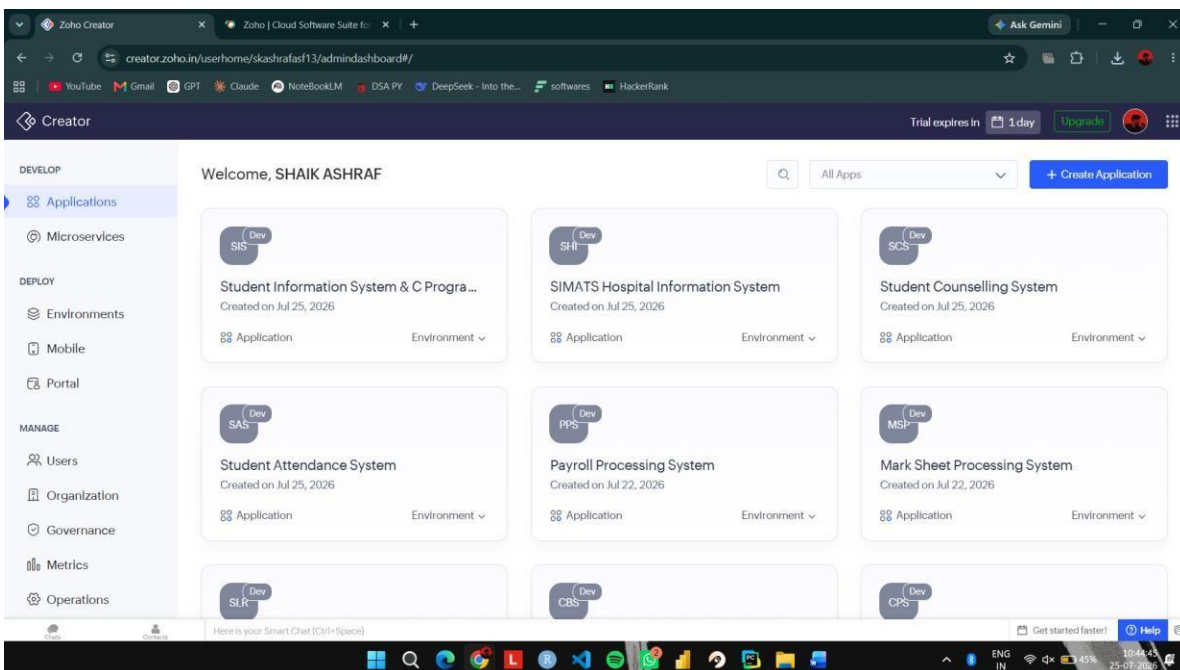
Step 6: Save and publish the application. The software has been created successfully.

IMPLEMENTATION:

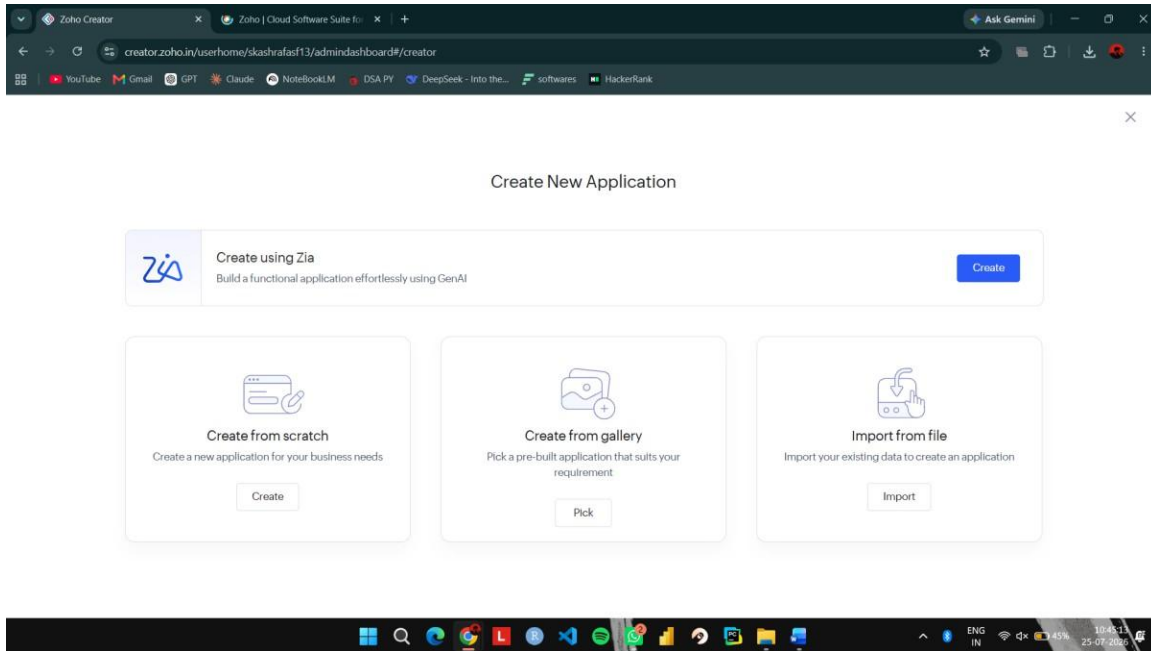
Step 1: Go to www.zoho.com



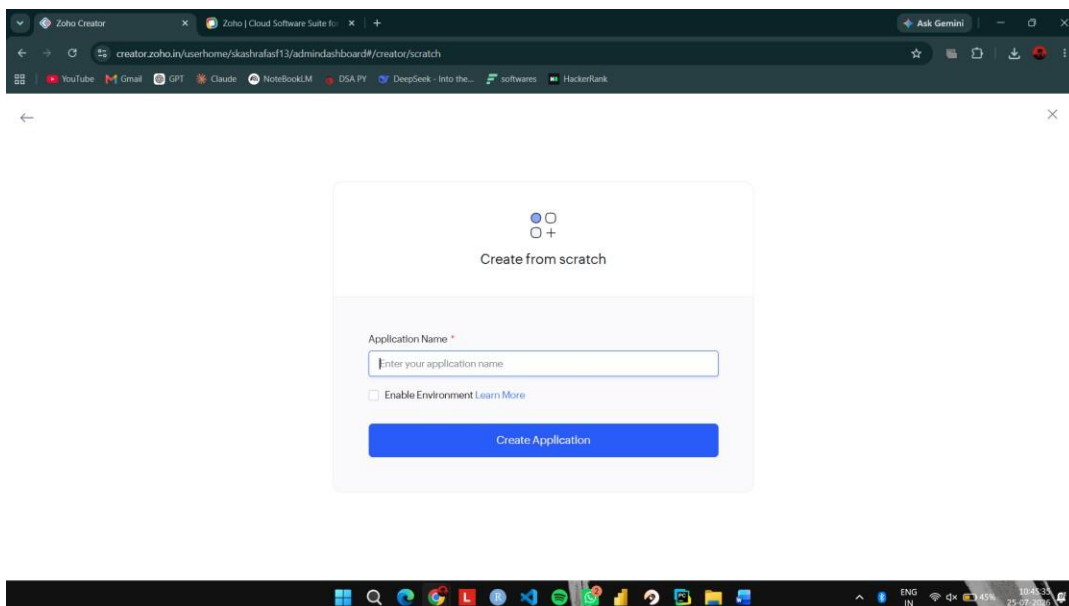
Step 2: Log in to your Zoho account.



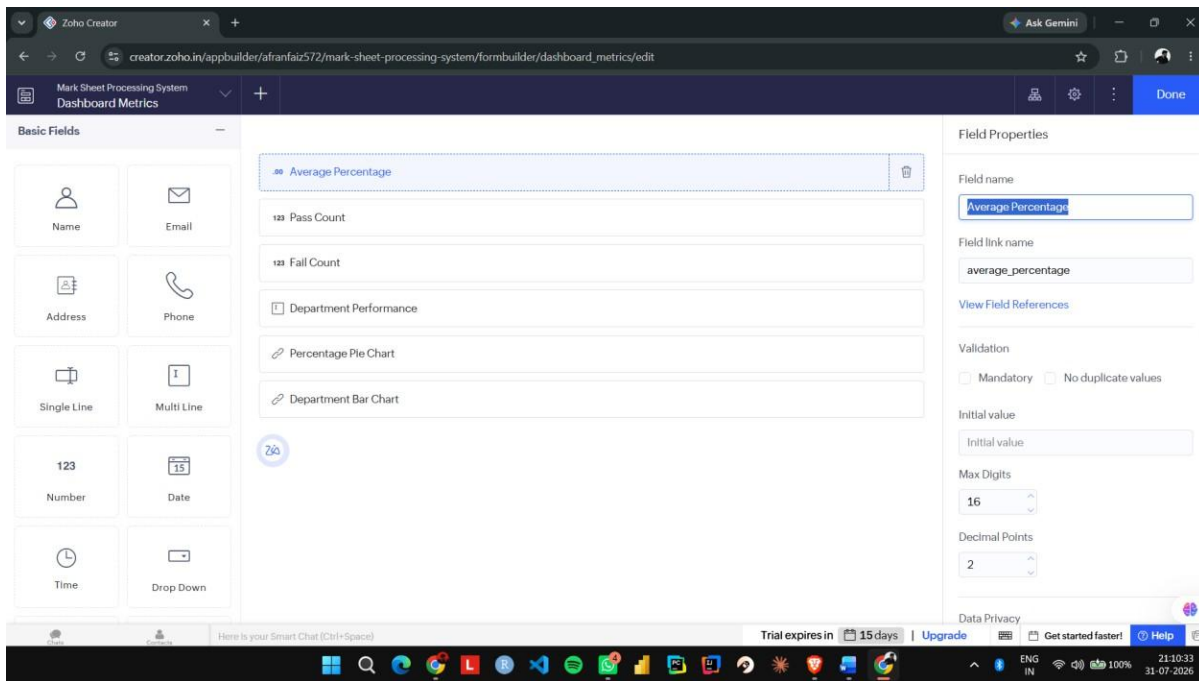
Step 3: Open Zoho Creator and select Create New Application.



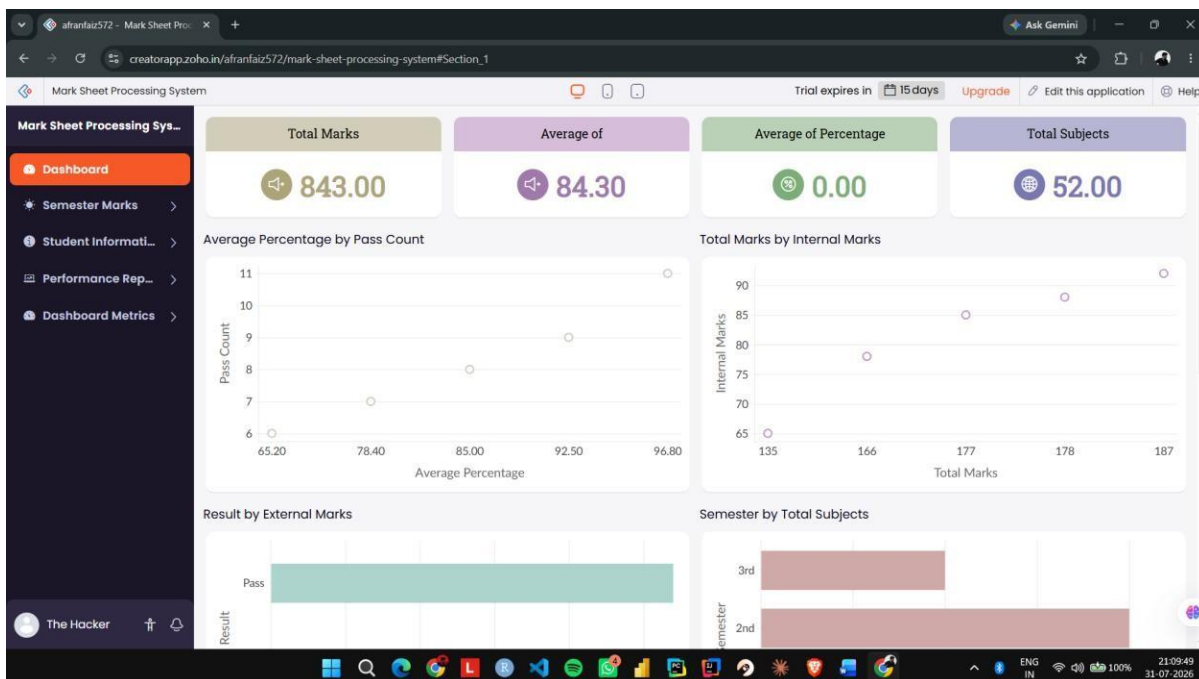
Step 4: Enter the application name as Mark Sheet Processing System.



Step 5: Create the required forms, reports, and dashboard.



Step 6: Save and publish the application. The software has been created successfully.



Experiment 3: Create a Clone of a VM and test it by loading the previous version/cloned VM.

Aim:

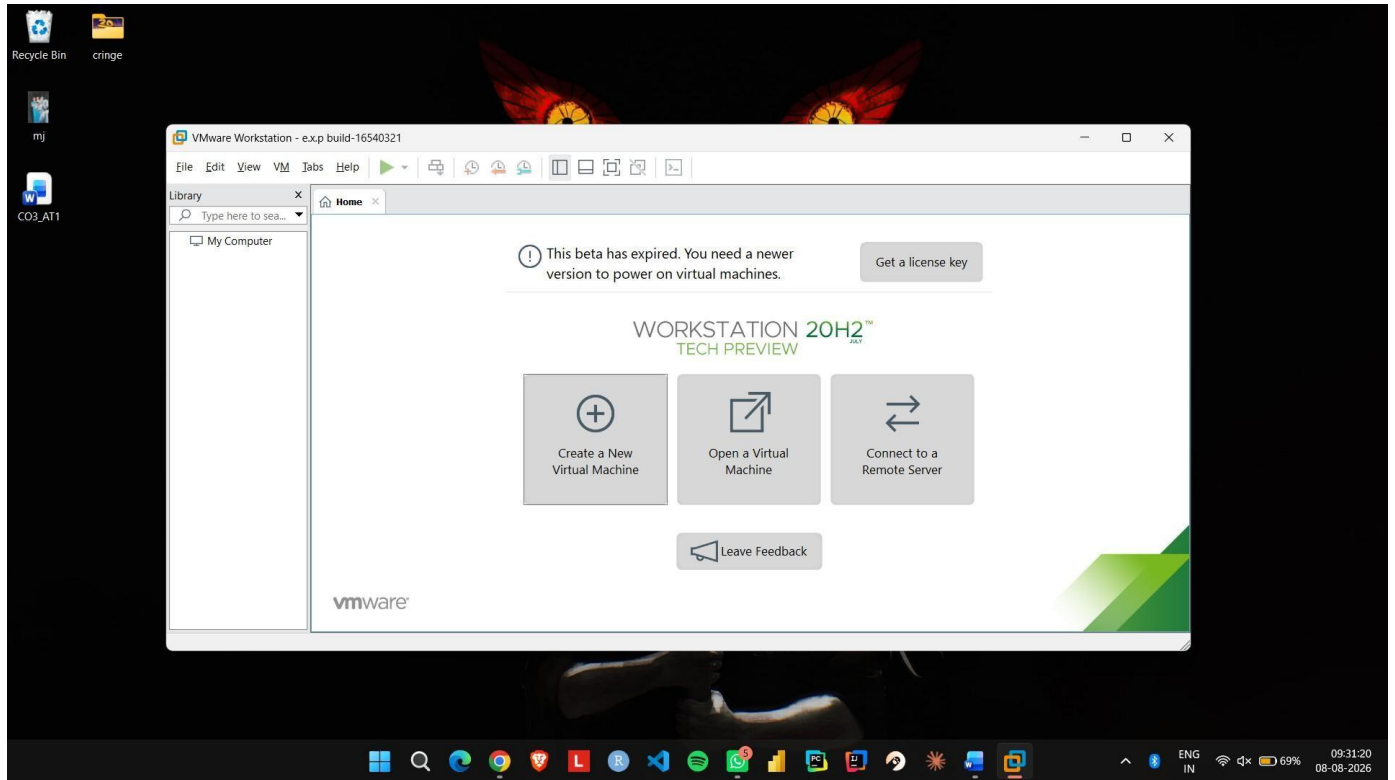
To create a clone of an existing virtual machine and test the cloned VM by restoring its previous version.

Procedure:

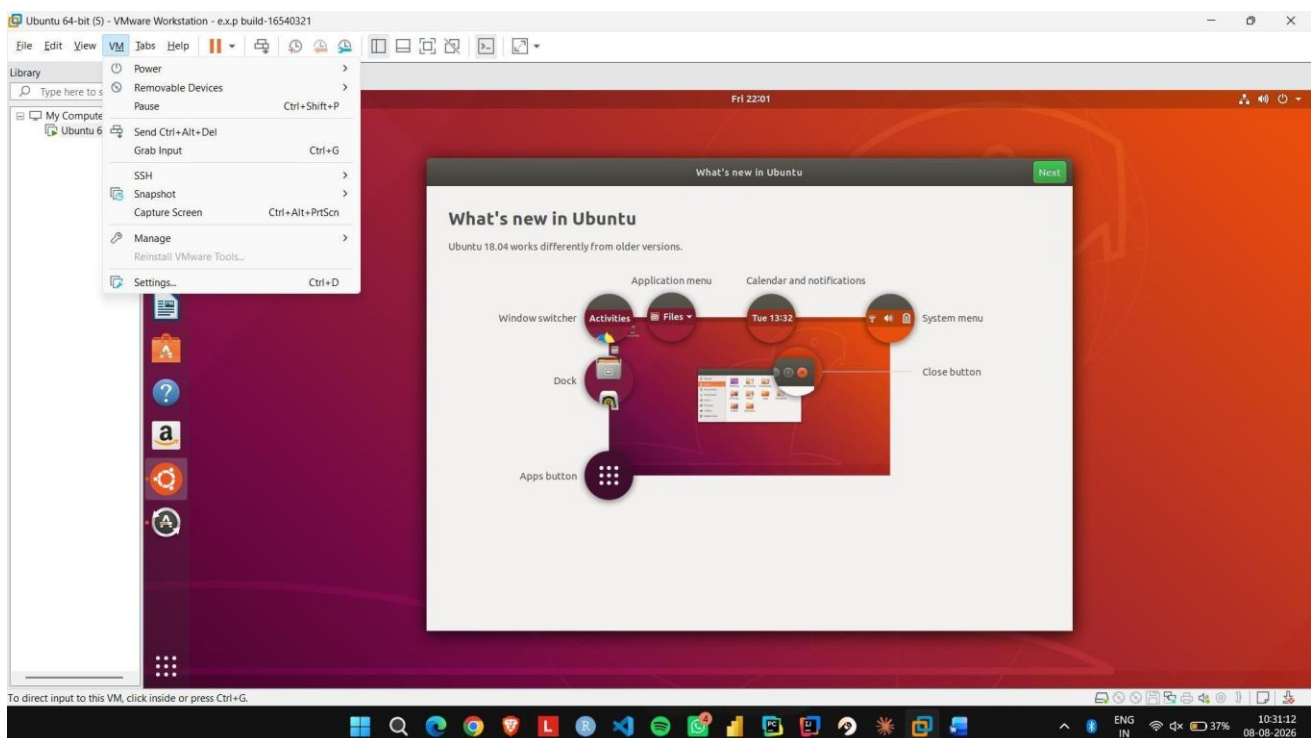
1. **STEP 1:** Open VMware Workstation, select the Ubuntu VM, and shut it down.
2. **STEP 2:** Select Manage → Clone to start the cloning process.
3. **STEP 3:** Select Full Clone and enter a name for the cloned VM.
4. **STEP 4:** Complete the cloning process and start the cloned VM.
5. **STEP 5:** Create a snapshot, make some changes, and then shut down the VM.
6. **STEP 6:** Restore the previous snapshot, start the VM, and verify that the previous state is loaded.

IMPLEMENTATION:

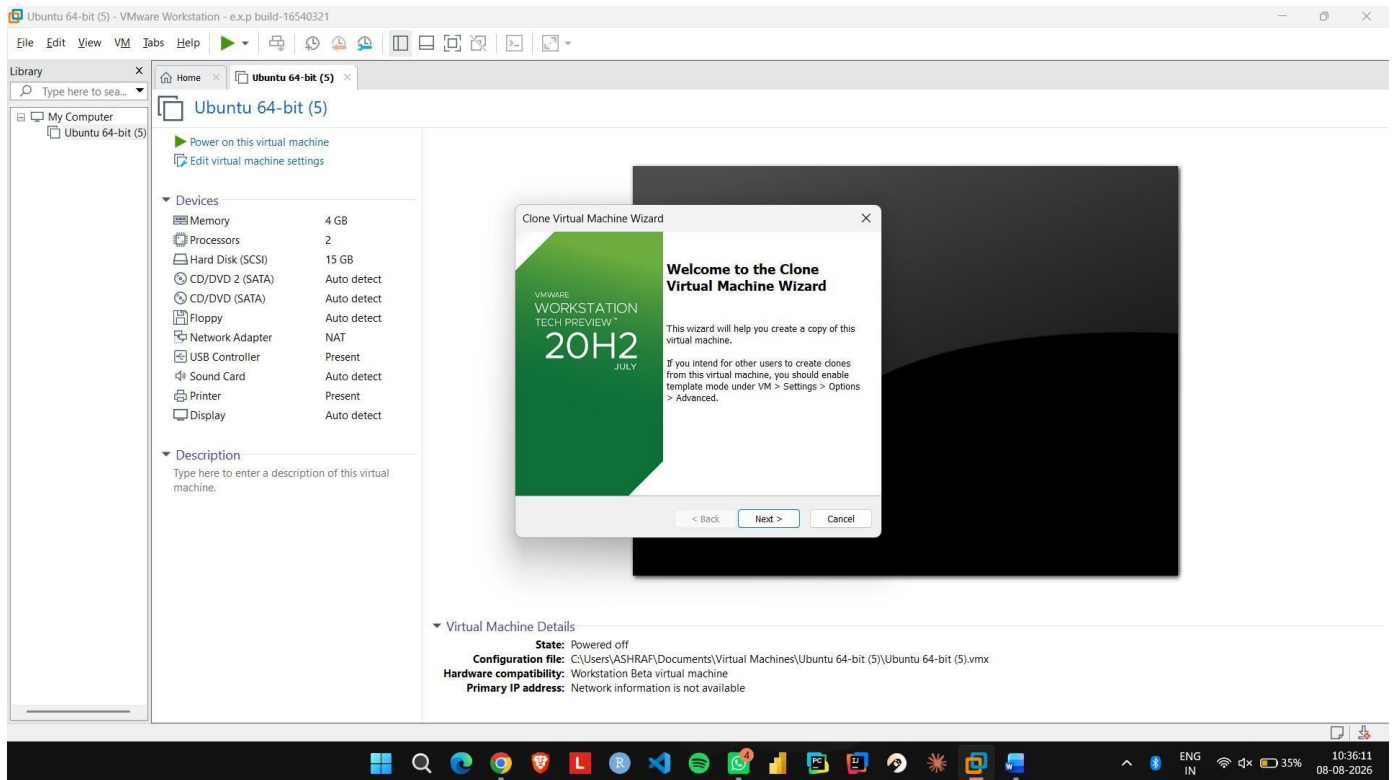
STEP 1: Open VMware Workstation, select the Ubuntu VM, and shut it down.



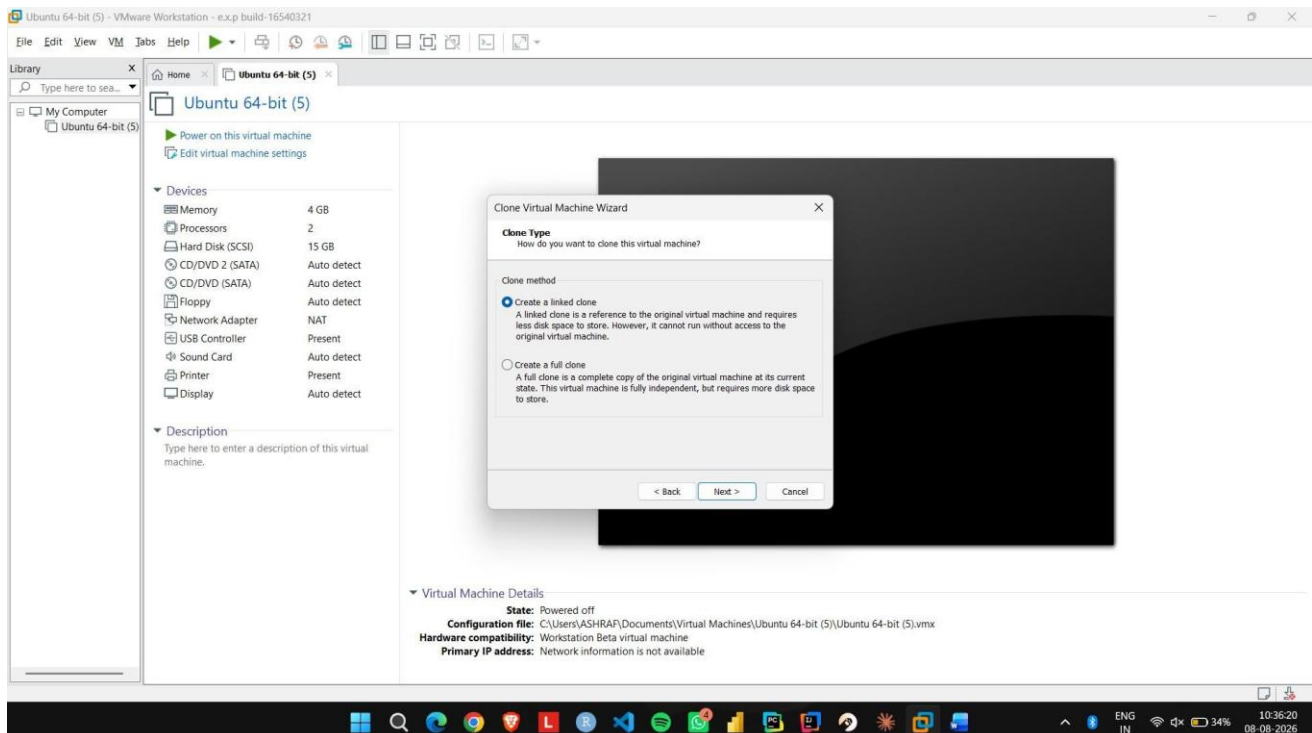
STEP 2: Select Manage → Clone to start the cloning process.



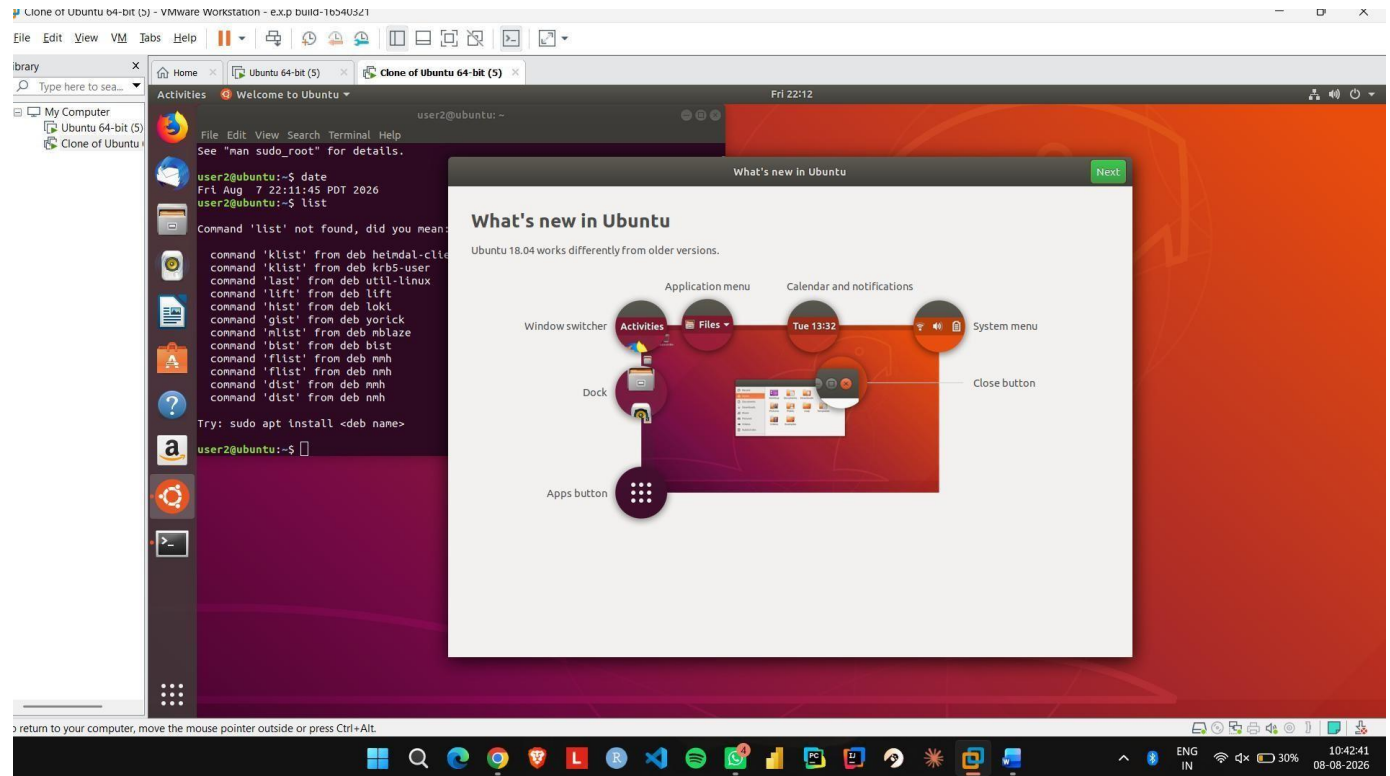
STEP 3: Select Full Clone and enter a name for the cloned VM.



STEP 4: Complete the cloning process and start the cloned VM.



STEP 5: Restore the previous snapshot, start the VM, and verify that the previous state is loaded.



Result: The VM was successfully **cloned and tested**, and its previous saved state was successfully restored.

Experiment 4: Install a C compiler in the virtual machine created using VirtualBox/VMware Workstation/Player and execute simple C programs.

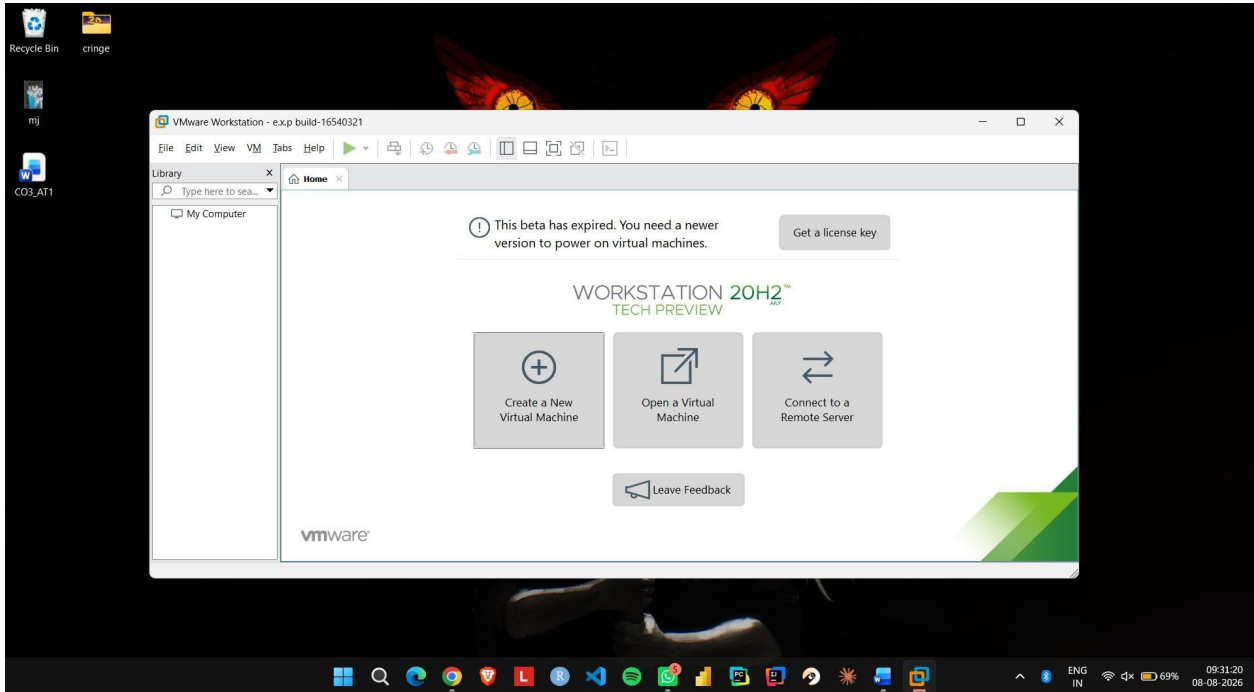
Aim: To install a C compiler (GCC) in a virtual machine created using VMware Workstation and execute simple C programs successfully.

Procedure:

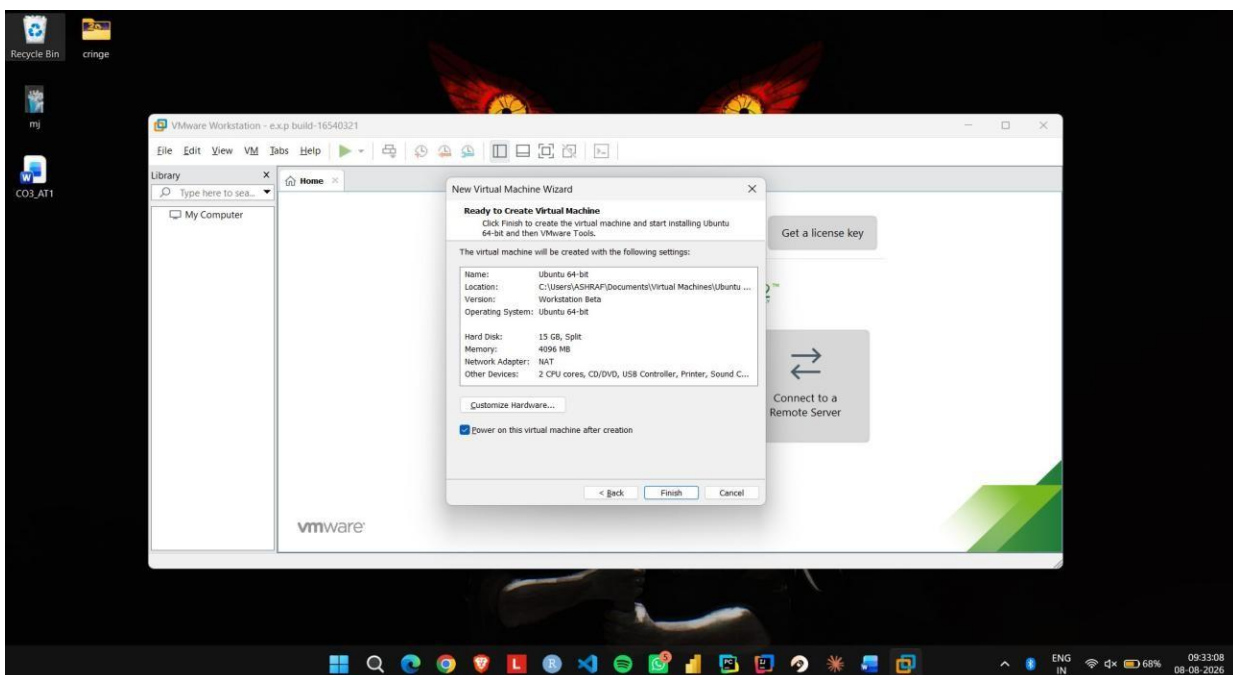
1. **STEP 1:** Open VMware Workstation and select Create a New Virtual Machine. Choose the Typical (Recommended) configuration and select the Ubuntu ISO file.
2. **STEP 2:** Set the Virtual Machine Name as Ubuntu-VM, configure RAM = 2 GB and CPU = 1 processor, and set the virtual hard disk size to 15 GB.
3. **STEP 3:** Start the Ubuntu-VM and complete the Ubuntu installation by following the on-screen instructions. After installation, log in to the Ubuntu operating system.
4. **STEP 4:** Open the Terminal and update the package information using `sudo apt update`. Then install the GCC C compiler using `sudo apt install gcc`.
5. **STEP 5:** Verify the compiler installation using `gcc --version`. Create a simple C program, save it with a .c extension, and compile it using `gcc program.c -o program`.
6. **STEP 6:** Execute the compiled program using `./program` and verify the output displayed in the terminal.

IMPLEMENTATION:

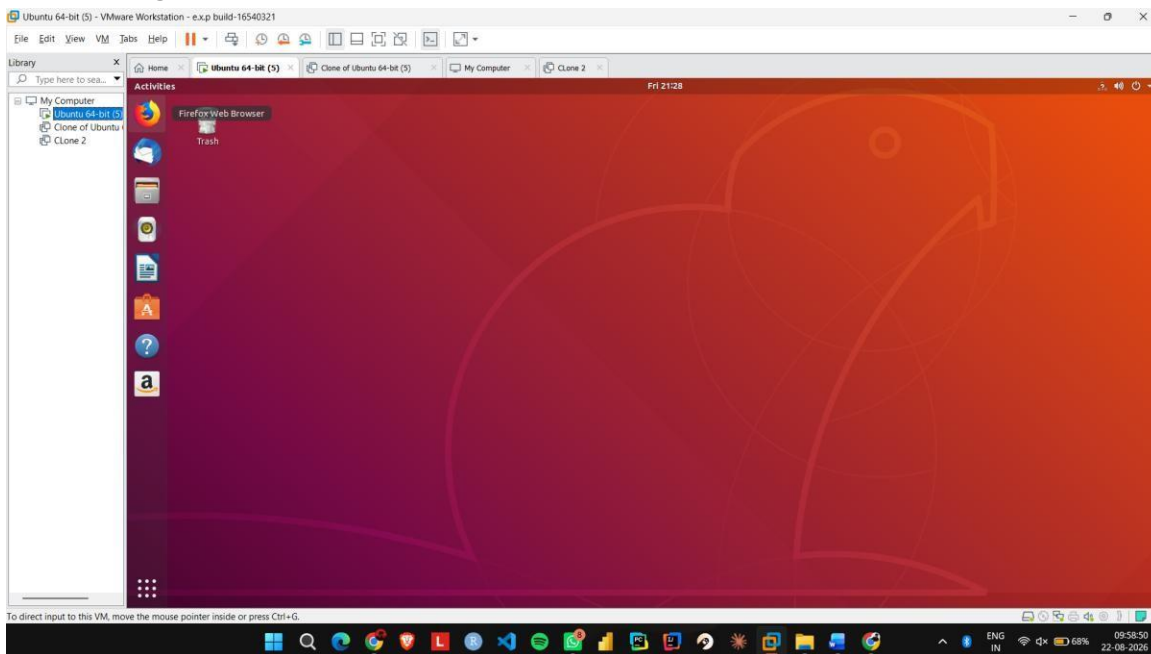
STEP 1: Open VMware Workstation and select Create a New Virtual Machine. Choose the configuration and select the Ubuntu ISO file.



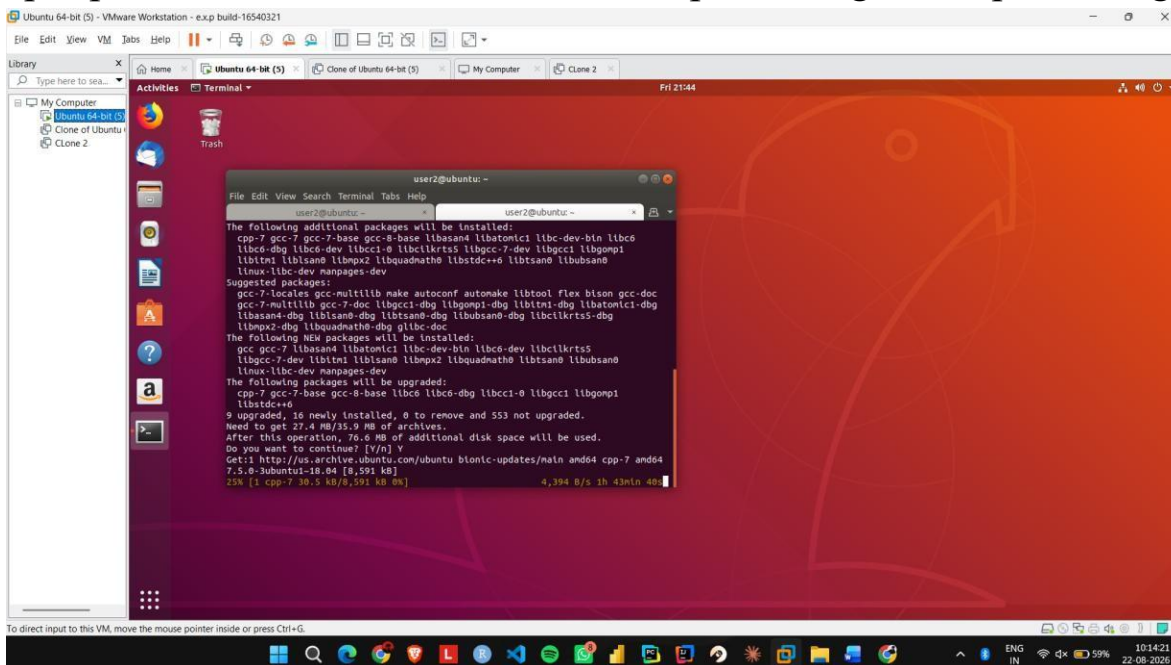
STEP 2: Set the Virtual Machine Name as Ubuntu-VM, configure RAM = 2 GB and CPU = 1 processor, and set the virtual hard disk size to 15 GB.



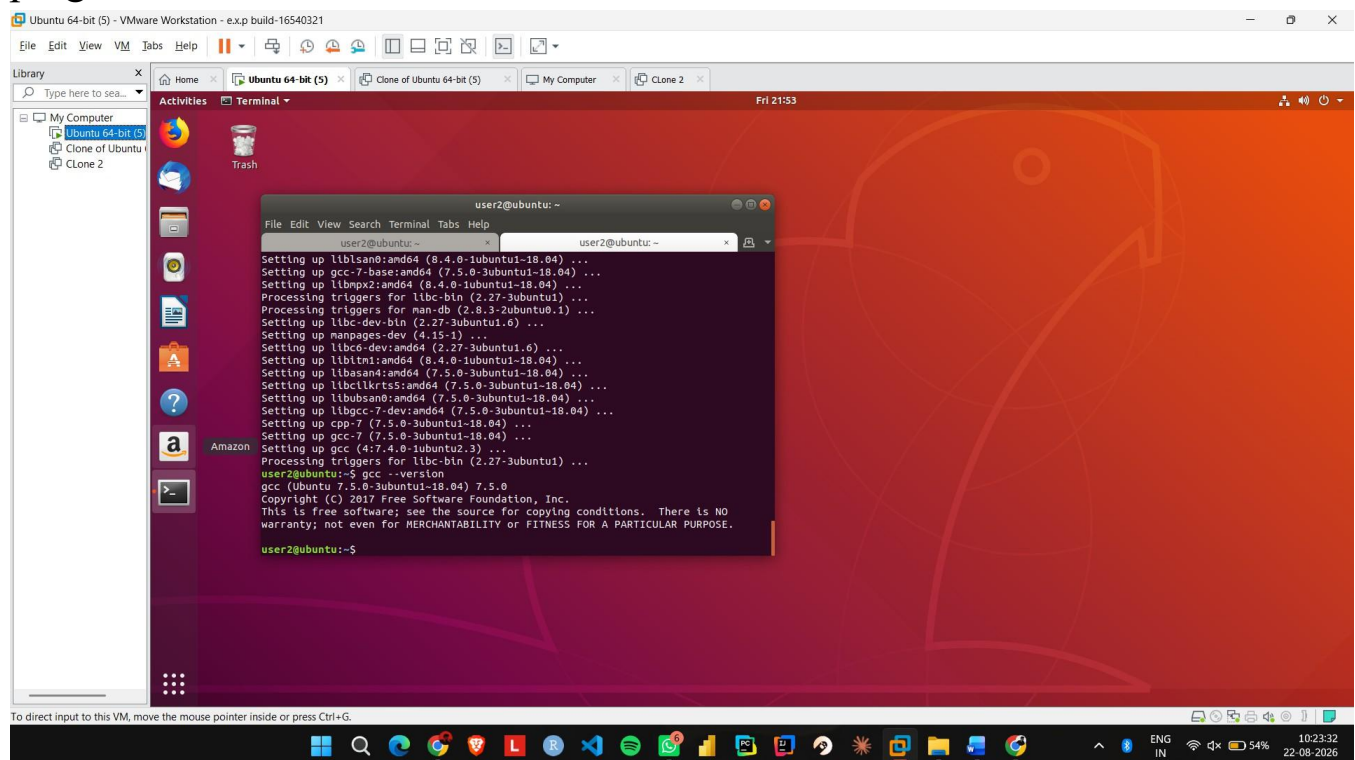
STEP 3: Start the Ubuntu-VM and complete the Ubuntu installation by following the on-screen instructions.



STEP 4: Open the Terminal and update the package information using `sudo apt update`. Then install the GCC C compiler using `sudo apt install gcc`.

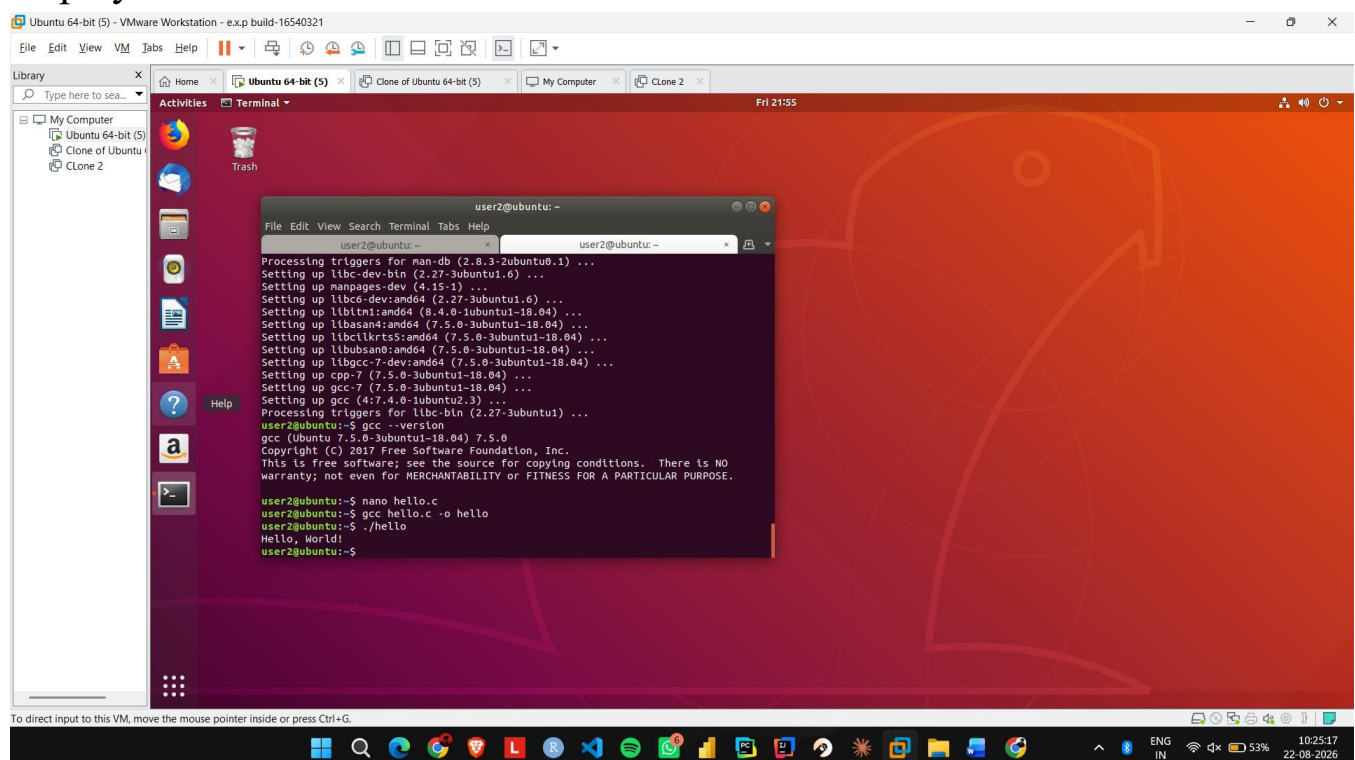


STEP 5: Verify the compiler installation using `gcc --version`. Create a simple C program, save it with a `.c` extension, and compile it using `gcc program.c -o program`.



```
user2@ubuntu: ~  
File Edit View Search Terminal Tabs Help  
user2@ubuntu: ~  
Setting up liblsan0:amd64 (8.4.0-1ubuntu1-18.04) ...  
Setting up gcc-7-base:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up libmpx2:amd64 (8.4.0-1ubuntu1-18.04) ...  
Processing triggers for libc-bin (2.27-3ubuntu1) ...  
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...  
Setting up libc-dev-bin (2.27-3ubuntu1.6) ...  
Setting up manpages-dev (4.15-1) ...  
Setting up libc6-dev:amd64 (2.27-3ubuntu1.6) ...  
Setting up libitm1:amd64 (8.4.0-1ubuntu1-18.04) ...  
Setting up libasan4:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up libcklkrts5:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up libubsan0:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up libgcc-7-dev:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up cpp-7 (7.5.0-3ubuntu1-18.04) ...  
Setting up gcc-7 (7.5.0-3ubuntu1-18.04) ...  
Setting up gcc (4:7.4.0-1ubuntu2.3) ...  
Processing triggers for libc-bin (2.27-3ubuntu1) ...  
user2@ubuntu:~$ gcc --version  
gcc (Ubuntu 7.5.0-3ubuntu1-18.04) 7.5.0  
Copyright (C) 2017 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
user2@ubuntu:~$
```

STEP 6: Execute the compiled program using `./program` and verify the output displayed in the terminal.



```
user2@ubuntu: ~  
File Edit View Search Terminal Tabs Help  
user2@ubuntu: ~  
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...  
Setting up libc-dev-bin (2.27-3ubuntu1.6) ...  
Setting up manpages-dev (4.15-1) ...  
Setting up libc6-dev:amd64 (2.27-3ubuntu1.6) ...  
Setting up libitm1:amd64 (8.4.0-1ubuntu1-18.04) ...  
Setting up libasan4:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up libcklkrts5:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up libubsan0:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up libgcc-7-dev:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up cpp-7 (7.5.0-3ubuntu1-18.04) ...  
Setting up gcc-7 (7.5.0-3ubuntu1-18.04) ...  
Setting up gcc (4:7.4.0-1ubuntu2.3) ...  
Processing triggers for libc-bin (2.27-3ubuntu1) ...  
user2@ubuntu:~$ gcc --version  
gcc (Ubuntu 7.5.0-3ubuntu1-18.04) 7.5.0  
Copyright (C) 2017 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
user2@ubuntu:~$ nano hello.c  
user2@ubuntu:~$ gcc hello.c -o hello  
user2@ubuntu:~$ ./hello  
Hello, World!  
user2@ubuntu:~$
```

Result: Thus, the GCC C compiler was successfully installed in the Ubuntu virtual machine created using VMware Workstation, and simple C programs were successfully compiled and executed.